

$$\begin{array}{ccccccc}
 \mathrm{Hom}(pA, \mathbb{Z}) & \longrightarrow & \mathrm{Ext}(A_p, \mathbb{Z}) & \longrightarrow & \mathrm{Ext}(A, \mathbb{Z}) & \xrightarrow{p} & \mathrm{Ext}(A, \mathbb{Z}) \longrightarrow \mathrm{Ext}(pA, \mathbb{Z}) \longrightarrow 0 \\
 & & & & \searrow & \nearrow & \\
 & & & & \mathrm{Ext}(pA, \mathbb{Z}) & & \\
 0 & \longrightarrow & \mathrm{Hom}(pA, \mathbb{Z}) & \xrightarrow{\approx} & \mathrm{Hom}(A, \mathbb{Z}) & \longrightarrow & 0
 \end{array}$$

where the map labeled ‘ p ’ is multiplication by p . Use this to show:

- (a) $\mathrm{Ext}(A, \mathbb{Z})$ is divisible iff A is torsionfree.
- (b) $\mathrm{Ext}(A, \mathbb{Z})$ is torsionfree if A is divisible, and the converse holds if $\mathrm{Hom}(A, \mathbb{Z}) = 0$.

3.G Transfer Homomorphisms

There is a simple construction called ‘transfer’ that provides very useful information about homology and cohomology of finite-sheeted covering spaces. After giving the definition and proving a few elementary properties, we will use the transfer in the construction of a number of spaces whose $\mathbb{Z}_p$ cohomology is a polynomial ring.

Let $\pi: \tilde{X} \rightarrow X$ be an n -sheeted covering space, for some finite n . In addition to the induced map on singular chains $\pi_\#: C_k(\tilde{X}) \rightarrow C_k(X)$ there is also a homomorphism in the opposite direction $\tau: C_k(X) \rightarrow C_k(\tilde{X})$ which assigns to a singular simplex $\sigma: \Delta^k \rightarrow X$ the sum of the n distinct lifts $\tilde{\sigma}: \Delta^k \rightarrow \tilde{X}$. This is obviously a chain map, commuting with boundary homomorphisms, so it induces **transfer homomorphisms** $\tau_*: H_k(X; G) \rightarrow H_k(\tilde{X}; G)$ and $\tau^*: H^k(\tilde{X}; G) \rightarrow H^k(X; G)$ for any coefficient group G . We focus on cohomology in what follows, but similar statements hold for homology as well.

The composition $\pi_\# \tau$ is clearly multiplication by n , hence $\tau^* \pi^* = n$. This has the consequence that the kernel of $\pi^*: H^k(X; G) \rightarrow H^k(\tilde{X}; G)$ consists of torsion elements of order dividing n , since $\pi^*(\alpha) = 0$ implies $\tau^* \pi^*(\alpha) = n\alpha = 0$. Thus the cohomology of $\tilde{X}$ must be ‘larger’ than that of X except possibly for torsion of order dividing n . This can be a genuine exception as one sees from the examples of S^m covering $\mathbb{R}P^m$ and lens spaces. More generally, if $S^m \rightarrow X$ is any n -sheeted covering space, then the relation $\tau^* \pi^* = n$ implies that $\tilde{H}^*(X; \mathbb{Z})$ consists entirely of torsion elements of order dividing n , apart from a possible $\mathbb{Z}$ in dimension m . (Since X is a closed manifold, its homology groups are finitely generated by Corollaries A.8 and A.9 in the Appendix.)

By studying the other composition $\pi^* \tau^*$ we will prove:

Proposition 3G.1. *Let $\pi: \tilde{X} \rightarrow X$ be an n -sheeted covering space defined by an action of a group Γ on $\tilde{X}$. Then with coefficients in a field F whose characteristic is 0 or a prime not dividing n , the map $\pi^*: H^k(X; F) \rightarrow H^k(\tilde{X}; F)$ is injective with image the subgroup $H^*(\tilde{X}; F)^\Gamma$ consisting of classes α such that $\gamma^*(\alpha) = \alpha$ for all $\gamma \in \Gamma$.*

Proof: We have already seen that elements of the kernel of π^* have finite order dividing n , so π^* is injective for the coefficient fields we are considering here. It remains to describe the image of π^* . Note first that $\tau\pi_*$ sends a singular simplex $\Delta^k \rightarrow \tilde{X}$ to the sum of all its images under the Γ -action. Hence $\pi^*\tau^*(\alpha) = \sum_{\gamma \in \Gamma} \gamma^*(\alpha)$ for $\alpha \in H^k(\tilde{X}; F)$. If α is fixed under the action of Γ on $H^k(\tilde{X}; F)$, the sum $\sum_{\gamma \in \Gamma} \gamma^*(\alpha)$ equals $n\alpha$, so if the coefficient field F has characteristic 0 or a prime not dividing n , we can write $\alpha = \pi^*\tau^*(\alpha/n)$ and thus α lies in the image of π^* . Conversely, since $\pi\gamma = \pi$ for all $\gamma \in \Gamma$, we have $\gamma^*\pi^*(\alpha) = \pi^*(\alpha)$ for all α , and so the image of π^* is contained in $H^*(\tilde{X}; F)^\Gamma$. $\square$

Example 3G.2. Let $X = S^1 \vee S^k$, $k > 1$, with $\tilde{X}$ the n -sheeted cover corresponding to the index n subgroup of $\pi_1(X)$, so $\tilde{X}$ is a circle with n S^k 's attached at equally spaced points around the circle. The deck transformation group $\mathbb{Z}_n$ acts by rotating the circle, permuting the S^k 's cyclically. Hence for any coefficient group G , the invariant cohomology $H^*(\tilde{X}; G)^{\mathbb{Z}_n}$ is all of H^0 and H^1 , plus a copy of G in dimension k , the cellular cohomology classes assigning the same element of G to each S^k . Thus $H^i(\tilde{X}; G)^{\mathbb{Z}_n}$ is exactly the image of π^* for $i = 0$ and k , while the image of π^* in dimension 1 is the subgroup $nH^1(\tilde{X}; G)$. Whether this equals $H^1(\tilde{X}; G)^{\mathbb{Z}_n}$ or not depends on G . For $G = \mathbb{Q}$ or $\mathbb{Z}_p$ with p not dividing n , we have equality, but not for $G = \mathbb{Z}$ or $\mathbb{Z}_p$ with p dividing n . In this last case the map π^* is not injective on H^1 .

Spaces with Polynomial mod p Cohomology

An interesting special case of the general problem of realizing graded commutative rings as cup product rings of spaces is the case of polynomial rings $\mathbb{Z}_p[x_1, \dots, x_n]$ over the coefficient field $\mathbb{Z}_p$, p prime. The basic question here is, which sets of numbers $d_1, \dots, d_n$ are realizable as the dimensions $|x_i|$ of the generators x_i ? From §3.2 we have the examples of products of $\mathbb{C}P^\infty$'s and $\mathbb{H}P^\infty$'s with d_i 's equal to 2 or 4, for arbitrary p , and when $p = 2$ we can also take $\mathbb{R}P^\infty$'s with d_i 's equal to 1.

As an application of transfer homomorphisms we will construct some examples with larger d_i 's. In the case of polynomials in one variable, it turns out that these examples realize everything that can be realized. But for two or more variables, more sophisticated techniques are necessary to realize all the realizable cases; see the end of this section for further remarks on this.

The construction can be outlined as follows. Start with a space Y already known to have polynomial cohomology $H^*(Y; \mathbb{Z}_p) = \mathbb{Z}_p[y_1, \dots, y_n]$, and suppose there is an action of a finite group Γ on Y . A simple trick called the Borel construction shows that without loss of generality we may assume the action is free, defining a covering space $Y \rightarrow Y/\Gamma$. Then by Proposition 3G.1 above, if p does not divide the order of Γ , $H^*(Y/\Gamma; \mathbb{Z}_p)$ is isomorphic to the subring of $\mathbb{Z}_p[y_1, \dots, y_n]$ consisting of polynomials that are invariant under the induced action of Γ on $H^*(Y; \mathbb{Z}_p)$. And in some cases this subring is itself a polynomial ring.

For example, if Y is the product of n copies of $\mathbb{CP}^\infty$ then the symmetric group Σ_n acts on Y by permuting the factors, with the induced action on $H^*(Y; \mathbb{Z}_p) \approx \mathbb{Z}_p[\gamma_1, \dots, \gamma_n]$ permuting the γ_i 's. A standard theorem in algebra says that the invariant polynomials form a polynomial ring $\mathbb{Z}_p[\sigma_1, \dots, \sigma_n]$ where σ_i is the i^{th} elementary symmetric polynomial, the sum of all products of i distinct γ_j 's. Thus σ_i is a homogeneous polynomial of degree i . The order of Σ_n is $n!$ so the condition that p not divide the order of Γ amounts to $p > n$. Thus we realize the polynomial ring $\mathbb{Z}_p[x_1, \dots, x_n]$ with $|x_i| = 2i$, provided that $p > n$.

This example is less than optimal since there happens to be another space, the Grassmann manifold of n -dimensional linear subspaces of $\mathbb{C}^\infty$, whose cohomology with any coefficient ring R is $R[x_1, \dots, x_n]$ with $|x_i| = 2i$, as we show in §4.D, so the restriction $p > n$ is not really necessary.

To get further examples the idea is to replace $\mathbb{CP}^\infty$ by a space with the same $\mathbb{Z}_p$ cohomology but with 'more symmetry,' allowing for larger groups Γ to act. The constructions will be made using $K(\pi, 1)$ spaces, which were introduced in §1.B. For a group π we constructed there a Δ -complex $B\pi$ with contractible universal cover $E\pi$. The construction is functorial: A homomorphism $\varphi: \pi \rightarrow \pi'$ induces a map $B\varphi: B\pi \rightarrow B\pi'$, $B\varphi([g_1 | \dots | g_n]) = [\varphi(g_1) | \dots | \varphi(g_n)]$, satisfying the functor properties $B(\varphi\psi) = B\varphi B\psi$ and $B\mathbb{1} = \mathbb{1}$. In particular, if Γ is a group of automorphisms of π , then Γ acts on $B\pi$.

The other ingredient we shall need is the **Borel construction**, which converts an action of a group Γ on a space Y into a free action of Γ on a homotopy equivalent space Y' . Namely, take $Y' = Y \times E\Gamma$ with the diagonal action of Γ , $\gamma(y, z) = (\gamma y, \gamma z)$ where Γ acts on $E\Gamma$ as deck transformations. The diagonal action is free, in fact a covering space action, since this is true for the action in the second coordinate. The orbit space of this diagonal action is denoted $Y \times_\Gamma E\Gamma$.

Example 3G.3. Let $\pi = \mathbb{Z}_p$ and let Γ be the full automorphism group $\text{Aut}(\mathbb{Z}_p)$. Automorphisms of $\mathbb{Z}_p$ have the form $x \mapsto mx$ for $(m, p) = 1$, so Γ is the multiplicative group of invertible elements in the field $\mathbb{Z}_p$. By elementary field theory this is a cyclic group, of order $p - 1$. The preceding constructions then give a covering space $K(\mathbb{Z}_p, 1) \rightarrow K(\mathbb{Z}_p, 1)/\Gamma$ with $H^*(K(\mathbb{Z}_p, 1)/\Gamma; \mathbb{Z}_p) \approx H^*(K(\mathbb{Z}_p, 1); \mathbb{Z}_p)^\Gamma$. We may assume we are in the nontrivial case $p > 2$. From the calculation of the cup product structure of lens spaces in Example 3.41 or Example 3E.2 we have $H^*(K(\mathbb{Z}_p, 1); \mathbb{Z}_p) \approx \Lambda_{\mathbb{Z}_p}[\alpha] \otimes \mathbb{Z}_p[\beta]$ with $|\alpha| = 1$ and $|\beta| = 2$, and we need to figure out how Γ acts on this cohomology ring.

Let $\gamma \in \Gamma$ be a generator, say $\gamma(x) = mx$. The induced action of γ on $\pi_1 K(\mathbb{Z}_p, 1)$ is also multiplication by m since we have taken $K(\mathbb{Z}_p, 1) = B\mathbb{Z}_p \times E\Gamma$ and γ takes an edge loop $[g]$ in $B\mathbb{Z}_p$ to $[\gamma(g)] = [mg]$. Hence γ acts on $H_1(K(\mathbb{Z}_p, 1); \mathbb{Z})$ by multiplication by m . It follows that $\gamma(\alpha) = m\alpha$ and $\gamma(\beta) = m\beta$ since $H^1(K(\mathbb{Z}_p, 1); \mathbb{Z}_p) \approx \text{Hom}(H_1(K(\mathbb{Z}_p, 1)), \mathbb{Z}_p)$ and $H^2(K(\mathbb{Z}_p, 1); \mathbb{Z}_p) \approx \text{Ext}(H_1(K(\mathbb{Z}_p, 1)), \mathbb{Z}_p)$, and it is a gen-

eral fact, following easily from the definitions, that multiplication by an integer m in an abelian group H induces multiplication by m in $\text{Hom}(H, G)$ and $\text{Ext}(H, G)$.

Thus $\gamma(\beta^k) = m^k \beta^k$ and $\gamma(\alpha\beta^k) = m^{k+1} \alpha\beta^k$. Since m was chosen to be a generator of the multiplicative group of invertible elements of $\mathbb{Z}_p$, it follows that the only elements of $H^*(K(\mathbb{Z}_p, 1); \mathbb{Z}_p)$ fixed by γ , hence by Γ , are the scalar multiples of $\beta^{i(p-1)}$ and $\alpha\beta^{i(p-1)-1}$. Thus $H^*(K(\mathbb{Z}_p, 1); \mathbb{Z}_p)^\Gamma = \Lambda_{\mathbb{Z}_p}[\alpha\beta^{p-2}] \otimes \mathbb{Z}_p[\beta^{p-1}]$, so we have produced a space whose $\mathbb{Z}_p$ cohomology ring is $\Lambda_{\mathbb{Z}_p}[x_{2p-3}] \otimes \mathbb{Z}_p[y_{2p-2}]$, subscripts indicating dimension.

Example 3G.4. As an easy generalization of the preceding example, replace the group Γ there by a subgroup of $\text{Aut}(\mathbb{Z}_p)$ of order d , where d is any divisor of $p-1$. The new Γ is generated by the automorphism $x \mapsto m^{(p-1)/d} x$, and the same analysis shows that we obtain a space with $\mathbb{Z}_p$ cohomology $\Lambda_{\mathbb{Z}_p}[x_{2d-1}] \otimes \mathbb{Z}_p[y_{2d}]$, subscripts again denoting dimension. For a given choice of d the condition that d divides $p-1$ says $p \equiv 1 \pmod{d}$, which is satisfied by infinitely many p 's, according to a classical theorem of Dirichlet.

Example 3G.5. The two preceding examples can be modified so as to eliminate the exterior algebra factors, by replacing $\mathbb{Z}_p$ by $\mathbb{Z}_{p^\infty}$, the union of the increasing sequence $\mathbb{Z}_p \subset \mathbb{Z}_{p^2} \subset \mathbb{Z}_{p^3} \subset \cdots$. The first step is to show that $H^*(K(\mathbb{Z}_{p^\infty}, 1); \mathbb{Z}_p) \approx \mathbb{Z}_p[\beta]$ with $|\beta| = 2$. We know that $\tilde{H}_*(K(\mathbb{Z}_{p^i}, 1); \mathbb{Z})$ consists of $\mathbb{Z}_{p^i}$'s in odd dimensions. The inclusion $\mathbb{Z}_{p^i} \hookrightarrow \mathbb{Z}_{p^{i+1}}$ induces a map $K(\mathbb{Z}_{p^i}, 1) \rightarrow K(\mathbb{Z}_{p^{i+1}}, 1)$ that is unique up to homotopy. We can take this map to be a p -sheeted covering space since the covering space of a $K(\mathbb{Z}_{p^{i+1}}, 1)$ corresponding to the unique index p subgroup of $\pi_1 K(\mathbb{Z}_{p^{i+1}}, 1)$ is a $K(\mathbb{Z}_{p^i}, 1)$. The homology transfer formula $\pi_* \tau_* = p$ shows that the image of the induced map $H_n(K(\mathbb{Z}_{p^i}, 1); \mathbb{Z}) \rightarrow H_n(K(\mathbb{Z}_{p^{i+1}}, 1); \mathbb{Z})$ for n odd contains the multiples of p , hence this map is the inclusion $\mathbb{Z}_{p^i} \hookrightarrow \mathbb{Z}_{p^{i+1}}$. We can use the universal coefficient theorem to compute the induced map $H^*(K(\mathbb{Z}_{p^{i+1}}, 1); \mathbb{Z}_p) \rightarrow H^*(K(\mathbb{Z}_{p^i}, 1); \mathbb{Z}_p)$. Namely, the inclusion $\mathbb{Z}_{p^i} \hookrightarrow \mathbb{Z}_{p^{i+1}}$ induces the trivial map $\text{Hom}(\mathbb{Z}_{p^{i+1}}, \mathbb{Z}_p) \rightarrow \text{Hom}(\mathbb{Z}_{p^i}, \mathbb{Z}_p)$, so on odd-dimensional cohomology the induced map is trivial. On the other hand, the induced map on even-dimensional cohomology is an isomorphism since the map of free resolutions

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{p^i} & \mathbb{Z} & \longrightarrow & \mathbb{Z}_{p^i} \longrightarrow 0 \\ & & \downarrow \mathbb{1} & & \downarrow p & & \downarrow p \\ 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{p^{i+1}} & \mathbb{Z} & \longrightarrow & \mathbb{Z}_{p^{i+1}} \longrightarrow 0 \end{array}$$

dualizes to

$$\begin{array}{ccccccc} 0 & \longleftarrow & \text{Ext}(\mathbb{Z}_{p^i}, \mathbb{Z}_p) & \longleftarrow & \text{Hom}(\mathbb{Z}, \mathbb{Z}_p) & \xleftarrow{0} & \text{Hom}(\mathbb{Z}, \mathbb{Z}_p) \\ & & \uparrow & & \uparrow \mathbb{1} & & \uparrow \\ 0 & \longleftarrow & \text{Ext}(\mathbb{Z}_{p^{i+1}}, \mathbb{Z}_p) & \longleftarrow & \text{Hom}(\mathbb{Z}, \mathbb{Z}_p) & \xleftarrow{0} & \text{Hom}(\mathbb{Z}, \mathbb{Z}_p) \end{array}$$

Since $\mathbb{Z}_{p^\infty}$ is the union of the increasing sequence of subgroups $\mathbb{Z}_{p^i}$, the space $B\mathbb{Z}_{p^\infty}$ is the union of the increasing sequence of subcomplexes $B\mathbb{Z}_{p^i}$. We can therefore apply

Proposition 3F.5 to conclude that $H^*(K(\mathbb{Z}_{p^\infty}, 1); \mathbb{Z}_p)$ is zero in odd dimensions, while in even dimensions the map $H^*(K(\mathbb{Z}_{p^\infty}, 1); \mathbb{Z}_p) \rightarrow H^*(K(\mathbb{Z}_p, 1); \mathbb{Z}_p)$ induced by the inclusion $\mathbb{Z}_p \hookrightarrow \mathbb{Z}_{p^\infty}$ is an isomorphism. Thus $H^*(K(\mathbb{Z}_{p^\infty}, 1); \mathbb{Z}_p) \approx \mathbb{Z}_p[\beta]$ as claimed.

Next we show that the map $\text{Aut}(\mathbb{Z}_{p^\infty}) \rightarrow \text{Aut}(\mathbb{Z}_p)$ obtained by restriction to the subgroup $\mathbb{Z}_p \subset \mathbb{Z}_{p^\infty}$ is a split surjection. Automorphisms of $\mathbb{Z}_{p^i}$ are the maps $x \mapsto mx$ for $(m, p) = 1$, so the restriction map $\text{Aut}(\mathbb{Z}_{p^{i+1}}) \rightarrow \text{Aut}(\mathbb{Z}_{p^i})$ is surjective. Since $\text{Aut}(\mathbb{Z}_{p^\infty}) = \varprojlim \text{Aut}(\mathbb{Z}_{p^i})$, the restriction map $\text{Aut}(\mathbb{Z}_{p^\infty}) \rightarrow \text{Aut}(\mathbb{Z}_p)$ is also surjective. The order of $\text{Aut}(\mathbb{Z}_{p^i})$, the multiplicative group of invertible elements of $\mathbb{Z}_{p^i}$, is $p^i - p^{i-1} = p^{i-1}(p-1)$ and $p-1$ is relatively prime to p^{i-1} , so the abelian group $\text{Aut}(\mathbb{Z}_{p^i})$ contains a subgroup of order $p-1$. This subgroup maps onto the cyclic group $\text{Aut}(\mathbb{Z}_p)$ of the same order, so $\text{Aut}(\mathbb{Z}_{p^i}) \rightarrow \text{Aut}(\mathbb{Z}_p)$ is a split surjection, hence so is $\text{Aut}(\mathbb{Z}_{p^\infty}) \rightarrow \text{Aut}(\mathbb{Z}_p)$.

Thus we have an action of $\Gamma = \text{Aut}(\mathbb{Z}_p)$ on $B\mathbb{Z}_{p^\infty}$ extending its natural action on $B\mathbb{Z}_p$. The Borel construction then gives an inclusion $B\mathbb{Z}_p \times_\Gamma E\Gamma \hookrightarrow B\mathbb{Z}_{p^\infty} \times_\Gamma E\Gamma$ inducing an isomorphism of $H^*(B\mathbb{Z}_{p^\infty} \times_\Gamma E\Gamma; \mathbb{Z}_p)$ onto the even-dimensional part of $H^*(B\mathbb{Z}_p \times_\Gamma E\Gamma; \mathbb{Z}_p)$, a polynomial algebra $\mathbb{Z}_p[\gamma_{2p-2}]$. Similarly, if d is any divisor of $p-1$, then taking Γ to be the subgroup of $\text{Aut}(\mathbb{Z}_p)$ of order d yields a space with $\mathbb{Z}_p$ cohomology the polynomial ring $\mathbb{Z}_p[\gamma_{2d}]$.

Example 3G.6. Now we enlarge the preceding example by taking products and bringing in the permutation group to produce a space with $\mathbb{Z}_p$ cohomology the polynomial ring $\mathbb{Z}_p[\gamma_{2d}, \gamma_{4d}, \dots, \gamma_{2nd}]$ where d is any divisor of $p-1$ and $p > n$. Let X be the product of n copies of $B\mathbb{Z}_{p^\infty}$ and let Γ be the group of homeomorphisms of X generated by permutations of the factors together with the actions of $\mathbb{Z}_d$ in each factor constructed in the preceding example. We can view Γ as a group of $n \times n$ matrices with entries in $\mathbb{Z}_p$, the matrices obtained by replacing some of the 1's in a permutation matrix by elements of $\mathbb{Z}_p$ of multiplicative order a divisor of d . Thus there is a split short exact sequence $0 \rightarrow (\mathbb{Z}_d)^n \rightarrow \Gamma \rightarrow \Sigma_n \rightarrow 0$, and the order of Γ is $d^n n!$. The product space X has $H^*(X; \mathbb{Z}_p) \approx \mathbb{Z}_p[\beta_1, \dots, \beta_n]$ with $|\beta_i| = 2$, so $H^*(X \times_\Gamma E\Gamma; \mathbb{Z}_p) \approx \mathbb{Z}_p[\beta_1, \dots, \beta_n]^\Gamma$ provided that p does not divide the order of Γ , which means $p > n$. For a polynomial to be invariant under the $\mathbb{Z}_d$ action in each factor it must be a polynomial in the powers β_i^d , and to be invariant under permutations of the variables it must be a symmetric polynomial in these powers. Since symmetric polynomials are exactly the polynomials in the elementary symmetric functions, the polynomials in the β_i 's invariant under Γ form a polynomial ring $\mathbb{Z}_p[\gamma_{2d}, \gamma_{4d}, \dots, \gamma_{2nd}]$ with γ_{2k} the sum of all products of k distinct powers β_i^d .

Example 3G.7. As a further variant on the preceding example, choose a divisor q of d and replace Γ by its subgroup consisting of matrices for which the product of the q^{th} powers of the nonzero entries is 1. This has the effect of enlarging the ring of polynomials invariant under the action, and it can be shown that the invariant

polynomials form a polynomial ring $\mathbb{Z}_p[\mathcal{Y}_{2d}, \mathcal{Y}_{4d}, \dots, \mathcal{Y}_{2(n-1)d}, \mathcal{Y}_{2nq}]$, with the last generator $\mathcal{Y}_{2nd}$ replaced by $\mathcal{Y}_{2nq} = \prod_i \beta_i^q$. For example, if $n = 2$ and $q = 1$ we obtain $\mathbb{Z}_p[\mathcal{Y}_4, \mathcal{Y}_{2d}]$ with $\mathcal{Y}_4 = \beta_1\beta_2$ and $\mathcal{Y}_{2d} = \beta_1^d + \beta_2^d$. The group Γ in this case happens to be isomorphic to the dihedral group of order $2d$.

General Remarks

The problem of realizing graded polynomial rings $\mathbb{Z}_p[\mathcal{Y}]$ in one variable as cup product rings of spaces was discussed in §3.2, and Example 3G.5 provides the remaining examples, showing that $|\mathcal{Y}|$ can be any even divisor of $2(p-1)$. In more variables the problem of realizing $\mathbb{Z}_p[\mathcal{Y}_1, \dots, \mathcal{Y}_n]$ with specified dimensions $|\mathcal{Y}_i|$ is more difficult, but has been solved for odd primes p . Here is a sketch of the answer.

Assuming that p is odd, the dimensions $|\mathcal{Y}_i|$ are even. Call the number $d_i = |\mathcal{Y}_i|/2$ the *degree* of $\mathcal{Y}_i$. In the examples above this was in fact the degree of $\mathcal{Y}_i$ as a polynomial in the 2-dimensional classes β_j invariant under the action of Γ . It was proved in [Dwyer, Miller, & Wilkerson 1992] that every realizable polynomial algebra $\mathbb{Z}_p[\mathcal{Y}_1, \dots, \mathcal{Y}_n]$ is the ring of invariant polynomials $\mathbb{Z}_p[\beta_1, \dots, \beta_n]^\Gamma$ for an action of some finite group Γ on $\mathbb{Z}_p[\beta_1, \dots, \beta_n]$, where $|\beta_i| = 2$. The basic examples, whose products yield all realizable polynomial algebras, can be divided into two categories. First there are classifying spaces of Lie groups, each of which realizes a polynomial algebra for all but finitely many primes p . These are listed in the following table.

Lie group	degrees	primes
S^1	1	all
$SU(n)$	$2, 3, \dots, n$	all
$Sp(n)$	$2, 4, \dots, 2n$	all
$SO(2k)$	$2, 4, \dots, 2k-2, k$	$p > 2$
G_2	2, 6	$p > 2$
F_4	2, 6, 8, 12	$p > 3$
E_6	2, 5, 6, 8, 9, 12	$p > 3$
E_7	2, 6, 8, 10, 12, 14	$p > 3$
E_8	2, 8, 12, 14, 18, 20, 24, 30	$p > 5$

The remaining examples have to be constructed by hand. They form two infinite families plus 30 sporadic exceptions shown in the table on the next page. The first row is the examples we have constructed, though our construction needed the extra condition that p not divide the order of the group Γ . For all entries in both tables the order of Γ , the group such that $\mathbb{Z}_p[\mathcal{Y}_1, \dots, \mathcal{Y}_n] = \mathbb{Z}_p[\beta_1, \dots, \beta_n]^\Gamma$, turns out to equal the product of the degrees. When p does not divide this order, the method we used for the first row can also be applied to give examples for all the other rows. In some cases the congruence conditions on p , which are needed in order for Γ to be a subgroup of $\text{Aut}(\mathbb{Z}_p^n) = GL_n(\mathbb{Z}_p)$, automatically imply that p does not divide the order of Γ . But when this is not the case a different construction of a space with the

desired cohomology is needed. To find out more about this the reader can begin by consulting [Kane 1988] and [Notbohm 1999].

degrees		primes	
$d, 2d, \dots, (n-1)d, nq$ with $q \mid d$		$p \equiv 1 \pmod{d}$ $p \equiv -1 \pmod{d}$	
degrees	primes	degrees	primes
4, 6	$p \equiv 1 \pmod{3}$	60, 60	$p \equiv 1 \pmod{60}$
6, 12	$p \equiv 1 \pmod{3}$	12, 30	$p \equiv 1, 4 \pmod{15}$
4, 12	$p \equiv 1 \pmod{12}$	12, 60	$p \equiv 1, 49 \pmod{60}$
12, 12	$p \equiv 1 \pmod{12}$	12, 20	$p \equiv 1, 9 \pmod{20}$
8, 12	$p \equiv 1 \pmod{4}$	2, 6, 10	$p \equiv 1, 4 \pmod{5}$
8, 24	$p \equiv 1 \pmod{8}$	4, 6, 14	$p \equiv 1, 2, 4 \pmod{7}$
12, 24	$p \equiv 1 \pmod{12}$	6, 9, 12	$p \equiv 1 \pmod{3}$
24, 24	$p \equiv 1 \pmod{24}$	6, 12, 18	$p \equiv 1 \pmod{3}$
6, 8	$p \equiv 1, 3 \pmod{8}$	6, 12, 30	$p \equiv 1, 4 \pmod{15}$
8, 12	$p \equiv 1 \pmod{8}$	4, 8, 12, 20	$p \equiv 1 \pmod{4}$
6, 24	$p \equiv 1, 19 \pmod{24}$	2, 12, 20, 30	$p \equiv 1, 4 \pmod{5}$
12, 24	$p \equiv 1 \pmod{24}$	8, 12, 20, 24	$p \equiv 1 \pmod{4}$
20, 30	$p \equiv 1 \pmod{5}$	12, 18, 24, 30	$p \equiv 1 \pmod{3}$
20, 60	$p \equiv 1 \pmod{20}$	4, 6, 10, 12, 18	$p \equiv 1 \pmod{3}$
30, 60	$p \equiv 1 \pmod{15}$	6, 12, 18, 24, 30, 42	$p \equiv 1 \pmod{3}$

For the prime 2 the realization problem is still not completely solved. The known examples are listed in the short table at the right, where again we give only the irreducible examples, which generate others by taking products. All but the last entry in the table arise from classifying spaces of Lie groups, as described in §4.D. The construction for the last entry is in [Dwyer & Wilkerson 1993].

Lie group	degrees
$O(1)$	1
$SO(n)$	$2, 3, \dots, n$
$SU(n)$	$4, 6, \dots, 2n$
$Sp(n)$	$4, 8, \dots, 4n$
—	$8, 12, 14, 15$

3.H Local Coefficients

Homology and cohomology with local coefficients are fancier versions of ordinary homology and cohomology that can be defined for nonsimply-connected spaces. In various situations these more refined homology and cohomology theories arise naturally and inevitably. For example, the only way to extend Poincaré duality with $\mathbb{Z}$ coefficients to nonorientable manifolds is to use local coefficients. In the overall scheme of algebraic topology, however, the role played by local coefficients is fairly small. Local coefficients bring an extra level of complication that one tries to avoid whenever possible. With this in mind, the goal of this section will not be to give a full exposition but rather just to sketch the main ideas, leaving the technical details for the interested reader to fill in.